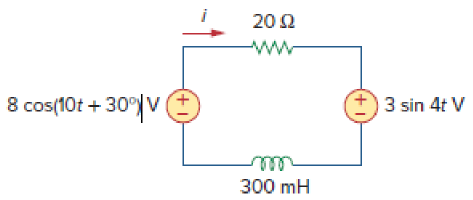
Chapter 10, Problem 45P

Problem

Use superposition to find *i*(*t*) in the circuit of Fig. 10.90.

Figure 10.90:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.90 in the textbook.

From Figure 10.90 in the textbook, the circuit operates at two different frequencies.

Consider the voltage source $8\cos(10t+30°)\,\mathrm{V}$ alone and short circuiting the voltage source $3\sin(4t)\,\mathrm{V}$.

Determine the value of inductive reactance.

$$\begin{aligned} X_L &= j\omega L \\ &= j(10)\left(300\times10^{-3}\right) \\ &= j3\ \Omega \end{aligned}$$

Re-draw the circuit diagram.

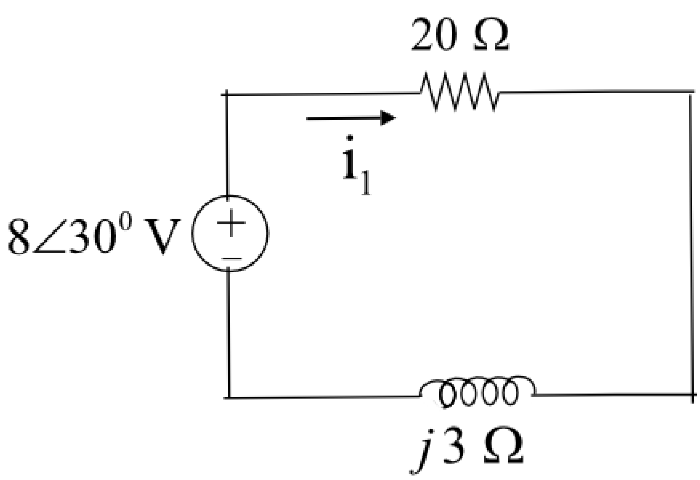


Figure 1

Step 2 of 7

Apply ohm's law to calculate the value of current, i_1.

$$i_1 = \frac{8\angle 30°}{20 + j3}$$
$$= \frac{8\angle 30°}{20.22\angle 8.53°}$$
$$= 0.3956\angle 21.47° \text{ A}$$

Convert the equation to time-domain form.

$$i_1(t) = 395.6\angle 21.47° \text{ mA}$$
$$= 395.6\cos(10t + 21.47°) \text{ mA}$$

Step 3 of 7

Consider the voltage source $3\sin(4t)$ V alone and short circuiting the voltage source $8\cos(10t + 30°)$ V.

Consider the source is, $V_s = 3\sin(4t)$ V.

Convert the source into cosine form.

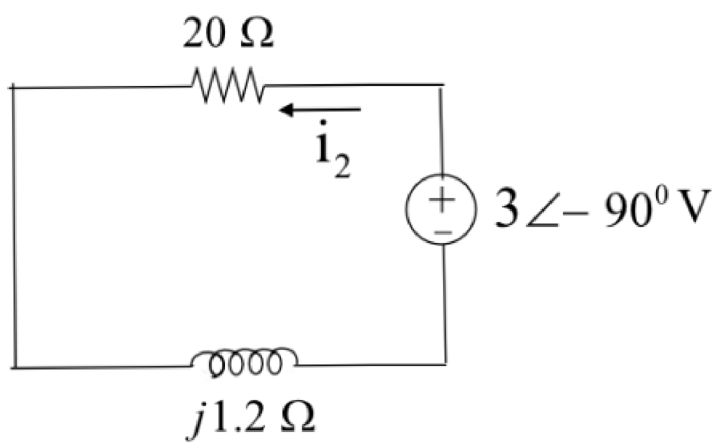
$$V_s = 3\sin(4t) \text{ V}$$
$$= 3\cos(4t - 90°) \text{ V}$$

Determine the value of inductive reactance.

$$X_L = j\omega L$$
$$= j(4)(300 \times 10^{-3})$$
$$= j1.2\ \Omega$$

Re-draw the circuit diagram.

Step 4 of 7



Step 5 of 7

Figure 2

Step 6 of 7

Apply ohm's law to calculate the value of current, i_2.

$$
\begin{aligned}
i_2 &= \frac{3\angle -90^\circ}{20 + j1.2} \\
&= \frac{20\angle -90^\circ}{20.04\angle 3.43^\circ} \\
&= 0.1497\angle -93.43^\circ \text{ A}
\end{aligned}
$$

Convert the equation to time-domain form.

$$
\begin{aligned}
i_2(t) &= 149.7\angle -93.43^\circ \text{ mA} \\
&= 149.7\cos(4t - 93.43^\circ) \text{ mA} \\
&= 149.7\sin(90^\circ + 4t - 93.43^\circ) \text{ mA} \\
&= 149.7\sin(4t - 3.43^\circ) \text{ mA}
\end{aligned}
$$

Step 7 of 7

Determine the value of current, $i(t)$.

$$
\begin{aligned}
i(t) &= i_1(t) - i_2(t) \\
&= 395.6\cos(10t + 21.47^\circ) \text{ mA} - 149.7\sin(4t - 3.43^\circ) \text{ mA} \\
&= 395.6\cos(10t + 21.47^\circ) \text{ mA} + 149.7\sin(180^\circ + 4t - 3.43^\circ) \text{ mA} \\
&= 395.6\cos(10t + 21.47^\circ) \text{ mA} + 149.7\sin(4t + 176.57^\circ) \text{ mA}
\end{aligned}
$$

Therefore, the value of current '$i(t)$' is

$\boxed{[395.6\cos(10t + 21.47^\circ) + 149.7\sin(4t + 176.57^\circ)] \text{ mA}}$.